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SUPERSEDING
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FEDERAL SPECIFICATION

PINE TAR, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers the product generally known in the naval stores industry as retort pine tar, the dark viscous liquid obtained by distilling off a part of the volatile oils contained in the tar oil portion of the condensate obtained when pine wood is destructively distilled (carbonized) in retorts.

1.2 Classification.

1.2.1 *Grade.*—The pine tar covered by this specification shall be of one grade, generally known in the trade as "medium" pine tar.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

SS-R-406—Road and Paving Materials; Methods of Sampling and Testing.

PPP-B-585—Boxes, Wood, Wirebound.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-C-96—Cans, Metal, 28 Gage and Lighter.

PPP-D-729—Drums; Metal, 55-Gallon (for Shipment of Noncorrosive Material).

PPP-D-760—Drums and Pails, Metal (5 and 16.64 Gallon).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Specifications:

MIL–R–3567—Reconditioning of Drums, Metal, 55-Gallon and Cans, Gasoline, Military, 5-Gallon.

Military Standard:

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publications of the issues in effect on date of invitation for bids, form a part of this specification.

American Society for Testing Materials Standards:

ASTM Designation D–20 — Standard Method of Test for Distillation of Tars and Tar Products.

ASTM Designation D–88 — Standard Method of Test for Viscosity by Means of Saybolt Viscosimeter.

ASTM Designation D–856—Standard Methods of Sampling and Testing Pine Tars.

ASTM Designation E–1 — Specifications for ASTM Thermometers.

(Copies of ASTM publications may be obtained by purchase from American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

Uniform Freight Classification.

(Copies may be obtained from Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Quality.—The pine tar supplied under this specification shall be free from miscible or immiscible non-tarry foreign matter other than moisture.

3.2 Appearance.—The pine tar shall be free-flowing and of uniform consistency (see 4.3.1).

3.3 Color.—The tar shall have a bright dark red-brown color when viewed through a thin film on a glass surface (see 4.3.2).

3.4 Spot test.—When one or two drops of the tar are rubbed out between two sheets of white writing paper, the resulting spot shall have a uniform light golden brown color, with not more than a trace of dirt specks or other foreign matter remaining unabsorbed by the paper (see 4.3.2).

3.5 Odor.—The odor of the tar shall be slightly empyreumatic, not unpleasant, and characteristic of retort tar made from dry seasoned wood.

3.6 Other detail requirements are as follows:

	Minimum	*Maximum*
Specific gravity (25/15.5°C.)	1.055	1.075
Moisture (percent by weight) (occluded and separated water)	----	2.5
Loss on heating (percent by weight)	----	8.0
Viscosity, centipoises	1400	2500
Distillation (percent by volume):		
Distilled below 170°C. (incl. water)	----	5
Distilled below 200°C.	----	8
Distilled below 370°C.	75	----

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

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4.2 Sampling and inspection. — Unless otherwise specified, sampling and inspection shall be in accordance with methods 1011, 1021 and 1031 of Federal Test Method Standard No. 141.

4.3 Test procedures.

4.3.1 *Appearance.*—Note the color and appearance of the pine tar when viewed by transmitted light through a thin layer on a clear glass surface (as in a partly filled glass container). Record the freedom of flow and uniformity of consistency of the tar.

4.3.2 *Color.*—Place a drop of the tar on a sheet of white writing paper, and rub out under a second sheet of such paper. Note the color of the translucent spot, and the presence or absence of specks or blotches of foreign matter.

4.3.3 *Specific gravity.*—Fill a 25-ml. Hubbard weighing bottle, previously standardized with distilled water at the temperature of test, with the tar, taking care to eliminate all air bubbles when inserting the stopper. Carefully clean and dry the outside of the bottle, and weigh at room temperature, to the third decimal place (nearest milligram). Compute the specific gravity as the weight of the tar divided by the weight of the water. If the temperature of test is above or below 25° C. add or substract the factor 0.0004 for each degree that the temperature varies above or below 25° C. This is the specific gravity at 25/25° C. which is converted to the 25/15.5° C. basis by multiplying by the factor 0.998.

4.3.4 *Moisture.*—Weigh 200 g. of tar into a tared, 500-ml. short-neck, round-bottom flask with 24/40 ground glass joint. Add 100 ml. dry xylene or petroleum solvent of such boiling range that not more than 5 percent distills below 110° C., and not less than 90 percent distills below 150° C. Mix the liquids by swirling the flask. Attach a Bidwell and Sterling or Dean and Stark moisture trap, and an upright condenser. The trap shall have a graduated leg of 10 ml., graduated in 0.1 ml. subdivisions. Bring the contents of the flask to a boil and reflux slowly at first (100 drops per minute). When recovery of the water is almost complete increase the reflux rate and maintain at 200 drops per minute to the end of the test. After 1 hour refluxing remove the heat and allow the apparatus to cool. Loosen any droplets of water adhering to the condenser wall with a copper wire spiral and wash down into the flask with small portions of the solvent. Read the volume of water in the trap and calculate the percentage to one decimal place.

4.3.5 *Loss on heating.* — Weigh approximately 25.00 g. of tar into a tared 50-mm. porcelain evaporating dish. Place in an oven maintained at 105° ± 1° C. and heat for exactly 1 hour. Remove the dish to a desiccator to cool and reweigh. Compute the percentage loss in weight to one decimal place.

4.3.6 *Viscosity (Saybolt methods).* — Determine the viscosity of the tar, recorded in seconds, at 50° C., by means of a Saybolt Furol Viscosimeter, according to method No. 2131 of Federal Specifications SS–R–406 (see also ASTM method D88).

4.3.7 *Viscosity (alternate Stormer method).*—For this alternate viscosity test, use a standard type of Stormer viscosimeter provided with a cylindrical rotor, test cup with two side vanes and thermometer well, movable weight box adjusted to weigh 150 g., and a Stormer viscosity thermometer (65 mm. partial immersion), range 20° to 70° C. conforming to the requirements for Thermometer 49c of ASTM Standard Thermometer Specifications E–1.

Place the instrument on a table or shelf, to allow the movable weight to descend without obstruction through a distance of about 40 inches, in order to permit at least 125 revolutions of the rotor in the tar. Insert the shaft of the rotor into the revolving chuck as far as possible, and fasten with the set screw. Fill the cup with the tar to cover the

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vanes about 1 inch. Place the cup in the water bath, and adjust the position of the bath so that the depth of tar above the rotor is equal to the clearance below the rotor. Insert the thermometer into the well, and note the temperature. If above 30° C., add ice to the water bath, to bring the temperature below that point. Then with a small flame under the bath, bring the tar to the required temperature, at the same time stirring it with the rotor by raising and releasing the movable weight several times. When the temperature of the tar is stable at 30° ± 0.5° C., raise the weight to the pulley, and secure it by turning the brake head screw. With stopwatch in hand, release the brake. After eight or ten revolutions, clock the time (in seconds) for the rotor to make 100 revolutions. Repeat the test several times to obtain a mean value for the time elapsed for 100 revolutions of the rotor. The reading of the thermometer must remain uniform through the test.

4.3.8 *Absolute viscosity.*—By either of the preceding methods (4.3.6 or 4.3.7) or by means of any other approved type or design of viscosimeter, determine the absolute viscosity of the tar at 30° ± 0.5° C., computed and recorded in centipoises.

4.3.9 *Distillation.*—Determine the distillation range of the tar in accordance with method 2180 of Federal Specification SS–R–406. (The apparatus and test procedure are similar to those described in ASTM Method D 20.). Should the temperature start to fall before the thermometer reading has reached 370° C., record the temperature and discontinue the test. Report the total quantity distilled to that temperature.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing protection used are defined in Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—As specified (see 6.2) pine tar shall be packaged in 5-gallon drums conforming to Federal Specification PPP–D–760, type II, class 1; 55-gallon drums conforming to Federal Specification PPP–D–729; or 5-gallon square metal cans conforming to Federal Specification PPP–C–96, type V, class 4. When stated in the invitation for bids and the purchase contract, the pine tar may be packaged in 5- or 55-gallon drums reconditioned to meet the quality requirements of Military Specification MIL–R–3567.

5.1.2 *Level C.*—The pine tar in the quantity specified (see 6.2) shall be packaged in accordance with the manufacturers commercial practice.

5.2 Packing.

5.2.1 *Level A.*—The pine tar packaged in the square metal cans specified in 5.1.1 shall be overpacked in wirebound, wood-cleated plywood, or nailed wood boxes conforming to Federal Specifications PPP–B–585, PPP–B–601, overseas type; or PPP–B–621, class 2, respectively. Metal cans shall be packed one per box. Five- and 55-gallon drums shall require no overpacking.

5.2.2 *Level B.*—Packing shall be as specified in 5.2.1 except that boxes shall be domestic type.

5.2.3 *Level C.*—Pine tar packaged in accordance with 5.1.1 or 5.1.2 as specified (see 6.2) shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.*—Individual containers and the outer cases holding such containers shall show the correct designation and type

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of pine tar therein, as set forth in this specification, and also the net quantity of contents in terms of pounds avoirdupois and subdivisions thereof. In addition, the name of the contractor and the number of the contract shall also be shown on drums or the outer cases in which cans are overpacked.

5.3.2 *Military agencies.* — In addition to any special marking specified in the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Kinds of pine tar.—Genuine pine tar is the heavy, dark viscous liquid resulting when a portion of the volatile oils are removed from the tar oil distillate recovered during the dry destructive distillation (carbonization) of pine wood. Several grades are marketed, under the designations thin, medium, heavy, and extra heavy, so classified on the basis of viscosity and the quantity of volatile oil remaining in the tar. A product sometimes offered as pine tar, or as "tarol", obtained by blending rosin oils and residues derived from the destructive distillation of rosin is not acceptable under this specification.

6.2 Intended use.—The pine tar covered by this specification is intended primarily for testing and waterproofing rope, cordage, netting, and other marine supplies.

6.3 Ordering data. — Purchasers should exercise any desired options offered herein, and should specify the following in procurement documents:

(*a*) Number, title, and date of this specification.
(*b*) The kind of pine tar.
(*c*) The size and type of unit containers (see 5.1).
(*d*) The selection of the applicable levels of packaging and packing (section 5).

6.4 Transportation description.—Transportation description and minimum weight applicable to this commodity are:

Rail:

Tar, pine

Carload minimum weight 40,000 pounds.

Motor:

Tar, pine

Motor volume minimum weight 40,000 pounds.

6.5 Inspection.—The United States Department of Agriculture maintains a Federal naval stores inspection and grading service in the areas where pine tar is produced. An agency purchasing pine tar may, by consulting the Naval Stores Branch, Agricultural Marketing Service, U. S. Department of Agriculture, Washington 25, D. C., determine whether it appears practicable and possible, on shipments originating in such producing areas, to effect a saving by delegating authority to that Branch to have its inspectors make the initial sampling and inspection, under reimbursement from the purchasing agency.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:
Army—Chemical Corps
Navy—Bureau of Supplies & Accounts.

Copies of this specification may be purchased for 5 cents each.

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